Homework 10, Lab 8

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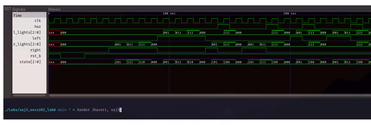

Figure 1: Waveform Output

```
../Labs/xaj3_eecs281_lab8 main ? ) iverilog -g2012 -o simulation.out testbench_hw8.sv tbird_fsm.sv

../Labs/xaj3_eecs281_lab8 main ? ) vvp simulation.out
VCD info: dumpfile dump.vcd opened for output.
rst_b | Left | Right | Haz | L_Lights | R_Lights
1     |  0   |  0    |  0   |   xxx    |   xxx
0     |  0   |  0    |  0   |   000    |   000
1     |  0   |  0    |  0   |   000    |   000
1     |  0   |  1    |  0   |   000    |   000
1     |  0   |  1    |  0   |   000    |   001
1     |  1   |  1    |  0   |   000    |   011
1     |  0   |  1    |  0   |   000    |   111
1     |  0   |  1    |  0   |   000    |   000
1     |  1   |  0    |  0   |   000    |   000
1     |  1   |  0    |  0   |   001    |   000
1     |  1   |  0    |  0   |   011    |   000
1     |  1   |  0    |  0   |   111    |   000
1     |  1   |  0    |  0   |   000    |   000
1     |  0   |  1    |  0   |   000    |   000
1     |  0   |  1    |  0   |   000    |   001
1     |  0   |  0    |  1   |   000    |   001
1     |  0   |  0    |  1   |   111    |   111
1     |  0   |  0    |  0   |   111    |   111
1     |  0   |  0    |  0   |   000    |   000
1     |  0   |  1    |  0   |   000    |   000
1     |  0   |  1    |  0   |   000    |   001
1     |  0   |  0    |  1   |   000    |   011
1     |  0   |  0    |  1   |   111    |   111
1     |  0   |  0    |  0   |   111    |   111
1     |  0   |  0    |  0   |   000    |   000
1     |  1   |  0    |  0   |   000    |   000
1     |  1   |  0    |  0   |   001    |   000
1     |  0   |  0    |  1   |   011    |   000
1     |  0   |  0    |  1   |   111    |   111
1     |  0   |  0    |  0   |   111    |   111
1     |  0   |  0    |  0   |   000    |   000
testbench_hw8.sv:66: $finish called at 270 (1s)

../Labs/xaj3_eecs281_lab8 main ? ) Xander Jhaveri, xaj3
```

Figure 2: Console Output

Note: I added some printing in the console just for testing purposes and for fun.

```
typedef enum logic [2:0] {
  IDLE = 3'b000,
  L1   = 3'b001,
  L2   = 3'b011,
  L3   = 3'b010,
  R1   = 3'b101,
  R2   = 3'b111,
  R3   = 3'b110,
  LR3  = 3'b100
} t_tbird_lights_state;

module tbird_fsm (
    input  logic       clk,
    input  logic       rst_b,
    input  logic       left,
    input  logic       right,
    input  logic       haz,
    output logic [2:0] l_lights,
    output logic [2:0] r_lights
);
  t_tbird_lights_state state;

  always_ff @(posedge clk or negedge rst_b) begin
    if (~rst_b) state <= IDLE;
    else
      case (state)
        IDLE: begin
          if ((left & right) | haz) state <= LR3;
          else if (left) state <= L1;
          else if (right) state <= R1;
          else state <= IDLE;
        end
        L1:  state <= t_tbird_lights_state'(haz ? LR3 : L2);
        L2:  state <= t_tbird_lights_state'(haz ? LR3 : L3);
        L3:  state <= IDLE;
        R1:  state <= t_tbird_lights_state'(haz ? LR3 : R2);
        R2:  state <= t_tbird_lights_state'(haz ? LR3 : R3);
        R3:  state <= IDLE;
        LR3: state <= IDLE;
      endcase
  end

  always_comb begin
    case (state)
      IDLE: begin
        l_lights = 3'b000;
        r_lights = 3'b000;
      end
      L1: begin
        l_lights = 3'b001;
        r_lights = 3'b000;
      end
      L2: begin
        l_lights = 3'b011;
        r_lights = 3'b000;
      end
      L3: begin
        l_lights = 3'b111;
        r_lights = 3'b000;
      end
      R1: begin
        l_lights = 3'b000;
        r_lights = 3'b001;
      end
      R2: begin
        l_lights = 3'b000;
        r_lights = 3'b011;
      end
      R3: begin
        l_lights = 3'b000;
        r_lights = 3'b111;
      end
      LR3: begin
        l_lights = 3'b111;
        r_lights = 3'b111;
      end
    endcase
  end

endmodule
```

Listing 1: `tbird_fsm.sv`

```
module testbench_hw8 ();

  logic clk, rst_b, left, right, haz;
  logic [2:0] l_lights, r_lights;

  tbird_fsm UUT (
      .clk(clk),
      .rst_b(rst_b),
      .left(left),
      .right(right),
      .haz(haz),
      .l_lights(l_lights),
      .r_lights(r_lights)
  );

  initial begin
    clk   = 1'b0;
    rst_b = 1'b1;
    left  = 1'b0;
    right = 1'b0;
    haz   = 1'b0;

    forever #5 clk = ~clk;
  end

  initial begin
    #10;
    rst_b = 1'b0;
    #20;
    rst_b = 1'b1;
    #20;
    right = 1'b1;
    #40;
    right = 1'b0;
    left  = 1'b1;
    #40;
    left  = 1'b0;
    right = 1'b1;
    #10;
    right = 1'b0;
    haz   = 1'b1;
    #10;
    haz = 1'b0;
    #10;
    right = 1'b1;
    #20;
    right = 1'b0;
    haz   = 1'b1;
    #10;
    haz = 1'b0;
    #10;
    left = 1'b1;
    #10;
    left = 1'b0;
    haz  = 1'b1;
    #10;
    haz = 1'b0;
    #10;
    left = 1'b1;
    #20;
    left = 1'b0;
    haz  = 1'b1;
    #10;
    haz = 1'b0;
    #10;
    $finish();
  end

  initial begin
    $dumpfile("dump.vcd");
    $dumpvars(0, testbench_hw8);

    $display("rst_b | Left | Right | Haz | L_Lights | R_Lights");
    $monitor("%1b     | %1b    | %1b     | %1b   |   %3b    |    %3b   ",
rst_b, left, right, haz,
              l_lights, r_lights);

    #280;
  end
endmodule
```

Listing 2: `testbench_hw8.sv`